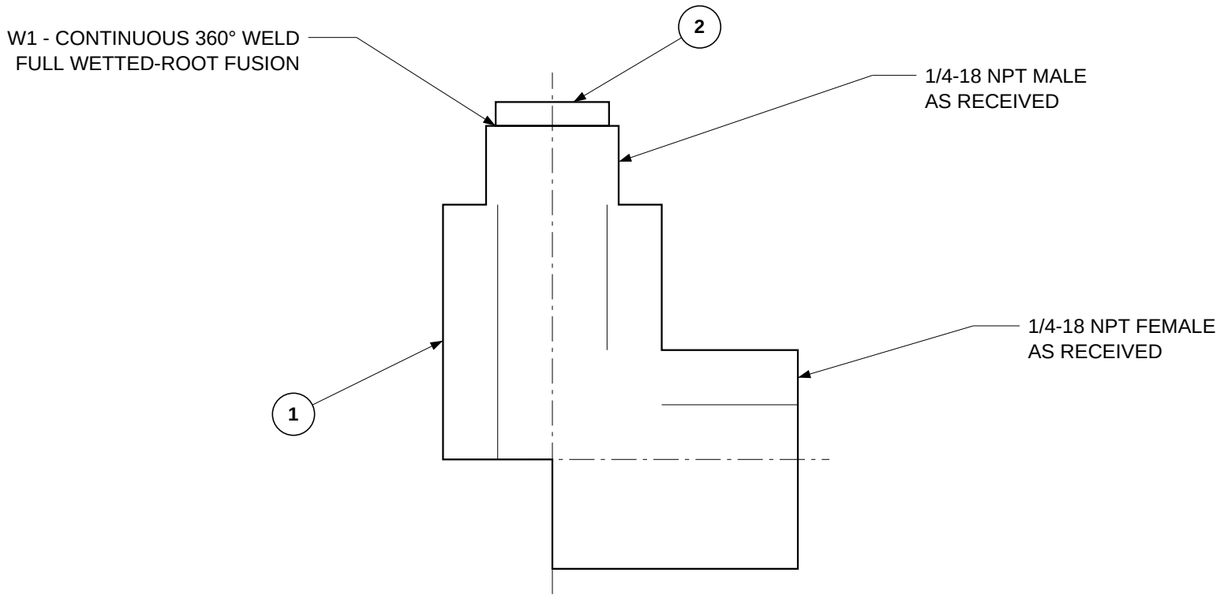
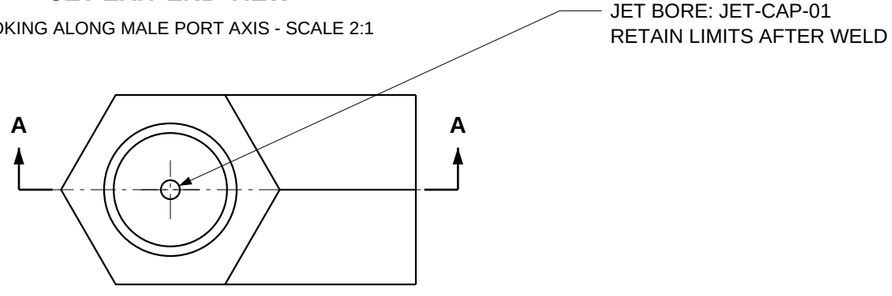
WATER-INLET JET ELBOW

SIDE VIEW - SCALE 2:1 - HIDDEN DETAIL OMITTED



JET EXIT END VIEW

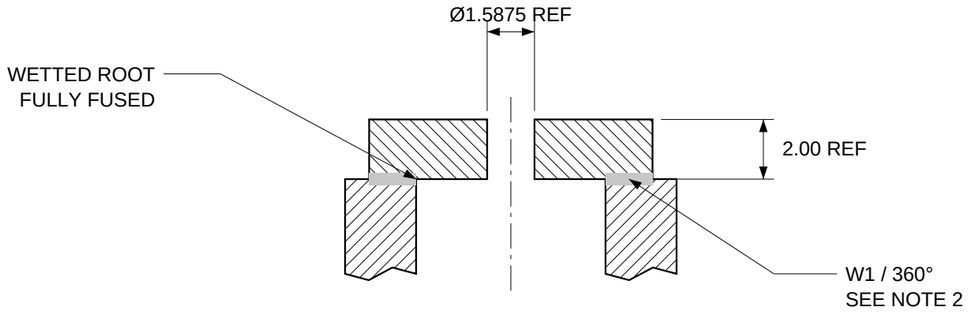
LOOKING ALONG MALE PORT AXIS - SCALE 2:1



FITTING BODY AND THREAD PROFILES ARE SCHEMATIC.

SECTION A-A - CAP-TO-ELBOW JOINT

LOCAL AT MALE TIP - SCALE 5:1



GREY BAND: REQUIRED FUSION PATH; WELD PROFILE DIAGRAMMATIC.

PARTS AT THIS WELD

ITEM	QTY	MATERIAL	DESCRIPTION / DEFINITION
1	1	HSM-0010	TAISHER 316L STREET ELBOW, 1/4 NPT M X F ORDER ID B0CZ38MYL1; AS RECEIVED
2	1	HSM-0011	316 STAINLESS DRILLED CAP; JET-CAP-01 FINISHED COMPONENT BEFORE WELDING

ASSEMBLY AND ACCEPTANCE

1. Assemble items 1 and 2. The purchased elbow controls fitting geometry. Body, bores and thread profiles are reference representations. Cap dimensions: JET-CAP-01.

2. W1: continuous 360° weld. Fuse the complete cap-to-tip interface through to the wetted bore-side root. Preserve the open jet. The grey band shows the required fusion path, not a weld size. Use the accepted small-joint coupon procedure; filler, if used: ER316L.

3. Cap OD axis within 0.10 radial of male-port bore axis; pre-weld seating gap 0.05 MAX. Completed jet axis within 0.20 radial of that bore axis and normal to the elbow tip within 1°. Retain JET-CAP-01 passage limits after welding.

4. Verify both threads after welding. Trial-fit through the loose END-CAP-01 plate at Port 2; the welded envelope must clear the actual tapped port and the jet exit must project into headspace. Record make-up depth, clocking and projection.

5. Inspect the cap and open passage; remove loose metal and cutting fluid. Flush, clean, passivate and flow-check the completed elbow per assembly/water-inlet-jet.md. Flow must emerge through the jet passage only.

6. Record actual bore/tip/land, cap overlap, maximum welded OD and port clearance. Qualify the same-material joint by sectioned-root inspection; retain settings and coupon identity. Numerical weld settings and flow acceptance require that qualification.

RELEASE ITEMS: MEASURED FIT / LAND AND PORT CLEARANCE; QUALIFIED SMALL-JOINT WELD; FLOW ACCEPTANCE.

HSM-0338

mm DIMENSIONS GOVERN. REF = REFERENCE.

PURCHASED FITTING CONTROLS ITS OWN GEOMETRY.

DO NOT SCALE. ANSI B / 17 x 11 IN. PRINT AT 100% FOR SCALE.

CAP COMPONENT: JET-CAP-01. ASSEMBLY: WATER-INLET-JET.MD.

HOME SODA MACHINE

CARBONATOR WATER-INLET JET ELBOW

ASSEMBLY DEFINITION

QUALIFICATION PENDING

MATERIAL	UNITS	QUANTITY
SEE PARTS LIST	**MILLIMETERS**	**1 PER CARBONATOR**

DRAWING	REV	DATE	SCALE	SHEET
JET-ELB-01	**A**	**2026-10-26**	**AS SHOWN**	**1 OF 1**